

IIIT-BGP IWSLT 2026 systems for Low-resource ST

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Abstract

We present low-resource Bhojpuri-Hindi speech translation systems for the IWSLT 2026 shared task, covering both end-to-end and cascaded settings. Our end-to-end model connects a Bhojpuri-finetuned Wav2Vec2 encoder to a pretrained NLLB-200 decoder via a lightweight interconnection adapter that combines learnable layer aggregation, CNN-based temporal compression, and Transformer refinement, with optional LoRA-based decoder adaptation. For our cascaded system, we finetune Whisper for Bhojpuri ASR and NLLB-200 for Hindi MT, and further apply QE Fusion with COMET-Kiwi to improve translation selection from beam candidates.

1 Introduction

Speech translation (ST) is the task of translating speech from one language into text in another, serving as an important component in breaking language barriers within communities (Joshi et al., 2019). While substantial progress has been made in ST for high-resource languages, performance in low-resource settings remains limited. One such scenario involves Bhojpuri, which, despite being spoken by over 50 million people, still falls into the low-resource category due to insufficient high-quality speech-to-text data, non-standardized orthography, and frequent code-mixing with Hindi/Angika, all of which complicate both speech recognition and translation.

ST systems are broadly categorized into two types:

- Cascaded
- End-to-end

Cascade speech translation systems, which combine automatic speech recognition (ASR)

and machine translation (MT), remain a common choice in low-resource settings due to the availability of pretrained ASR and MT models and their relatively low data requirements. However, such systems suffer from error propagation, where mistakes introduced during ASR are amplified by the downstream MT component, leading to degraded translation quality (Ruiz and Federico, 2014).

Recent end-to-end models, such as speech-to-text transformers, have shown remarkable performance in reducing ASR error propagation and the inherent latency of cascaded pipelines (Sperber and Paulik, 2020). However, these models typically require substantial labeled data, making them difficult to use for low-resource settings.

In this work, we systematically investigated both cascade and end-to-end approaches for low-resource Bhojpuri-Hindi speech translation. Our end-to-end system, combining learnable encoder-layer aggregation, a Transformer-based modality adapter, and Low-Rank Adaptation (LoRA) (Hu et al., 2022) based decoder adaptation, achieves a best dev BLEU of 32.77, while our cascade system with QE-guided hypothesis fusion yields consistent gains over the beam-search baseline across BLEU (Papineni et al., 2002), chrF++ (Popović, 2015), and COMET (Rei et al., 2020).

Our main contributions are:

- A modality adaptor combining CNN-based temporal compression with a Transformer refinement block to align Wav2Vec2 encoder representation with a pretrained NLLB decoder.
- Incorporation of an encoder-side, layer-wise aggregation strategy over intermediate Wav2Vec2 layers.

- Incorporation of parameter-efficient decoder adaptation using LoRA, comparing fully frozen, LoRA-only, and partially unfrozen configurations of the top decoder layers.
- QE Fusion (Vernikos and Popescu-Belis, 2024) as a replacement for standard beam search in the cascade MT component, using COMET-Kiwi (Rei et al., 2022) scores and reconstruct a superior hypothesis from 5 diverse candidates.

2 Related Work

ASR and MT have evolved from foundational linguistic tooling to sophisticated neural frameworks that leverage cross-lingual transfer and instruction tuning. Early computational efforts for Bhojpuri focused on establishing basic linguistic resources, such as the first statistical Part-of-Speech (POS) tagger using Support Vector Machines (Fernando et al., 2016). Mundotiya et al. (2021) extended this work by releasing annotated corpora covering POS, morphology, and chunking for Bhojpuri alongside related Purvanchal languages, providing the first systematic benchmarks for POS tagging, morphology analysis, and chunking. On the speech side, (Kumar et al., 2022) curated an annotated speech corpus for Bhojpuri and closely related varieties, establishing early ASR baselines. Neural MT for Bhojpuri-Hindi has received limited dedicated attention; the NLLB-200 project (Costa-Jussà et al., 2022) includes Bhojpuri as one of its languages, effectively making large-scale multilingual transfer the dominant strategy for this pair.

End-to-end speech translation (ST) directly maps source-language speech to target-language text without an explicit ASR-MT cascade. Early work showed that sequence-to-sequence models could directly perform speech translation, establishing the feasibility of direct ST (Weiss et al., 2017). More recent research has emphasized the importance of pretrained speech and text models, particularly in low-resource settings where parallel ST data is limited (Niehues et al., 2021). Another recent direction in end-to-end ST is to couple pretrained speech encoders with pretrained text decoders via lightweight adaptation modules. Prior work has shown that combining pretrained speech

models such as Wav2Vec 2.0 (Baevski et al., 2020a) or HuBERT with pretrained text decoders such as mBART can substantially improve end-to-end ST quality (Gállego et al., 2021; Tsiamas et al., 2022; Pham et al., 2022). In low-resource settings, related system papers have also explored tightly coupling pretrained ASR encoders with NLLB-based decoders (Avila and Crego, 2025).

Within this modular setting, two lines of work are particularly relevant to our end-to-end approach. M-Adapter (Zhao et al., 2022) addresses the modality gap between speech encoder outputs and text decoder inputs by introducing a dedicated adaptation module that compresses and transforms speech representations before decoding. Nishikawa and Nakamura (2023) shows that relying solely on the final speech encoder layer is suboptimal and instead aggregates intermediate encoder layers via learnable weights before passing them to the decoder.

In NMT, much attention has shifted towards neural quality estimators such as COMET (Rei et al., 2020), and its reference-free variant COMET-Kiwi (Rei et al., 2022), which score translation hypotheses without requiring reference translations and correlate more strongly with human judgments than surface-based metrics. Minimum Bayes Risk (MBR) decoding (Kumar and Byrne, 2004) offers one such approach: it selects the hypothesis with the highest expected utility, estimated against a set of sampled pseudo-references. Freitag et al. (Freitag et al., 2022) showed that replacing surface-based utility functions with neural metrics such as COMET yields substantial gains in human evaluation, establishing neural-metric MBR as a strong decoding baseline. Amongst other related works, QE reranking selects the top-scoring hypothesis from a candidate pool using a reference-free estimator, but is constrained to returning one of the existing candidates rather than constructing a new one (Rei et al., 2022). QE Fusion (Vernikos and Popescu-Belis, 2024) addresses this limitation by combining spans from multiple candidates. QE Fusion exploits inter-candidate complementarity to synthesize translations that outperform any single candidate, with consistent gains over beam search, QE reranking, and MBR across five language pairs, including on NLLB (Costa-Jussà et al.,

Dataset	Type	Train	Dev	Test	Total	Used In	
<i>Official Data (hours)</i>							
IWSLT 2026 Official	ST (bho→hi)	20h	2h	0.5h	22.5h	E2E	
<i>Additional Speech Data (hours)</i>							
HuggingFace ASR (real)	ASR (bho)	7h	0.6h	–	7.6h	Cascade	
HuggingFace ASR (synthetic)	ASR (bho)	4.2h	–	–	4.2h	Cascade	
<i>Additional Text Data (utterances)</i>							
Kaggle bho→hi Corpus	MT (bho→hi)	40.2k	4.5k	–	44.7k	Cascade	
Back-Translated	MT (bho→hi)	49.2k	5.5k	–	54.7k	Cascade	
<i>Combined Training Sets</i>							
Full Cascade Train Set	ASR+MT	89.5k / 11.2h		2h	0.5h	89.5k / 13.7h	Cascade
Full E2E Train Set	ST	40h	2h	0.5h	42h	E2E	

Table 1: Summary of datasets for Bhojpuri→Hindi ST. Speech in hours (h); text in thousands of utterances (k). Back-translated using facebook/nllb-200-1.3B (Costa-Jussà et al., 2022). The Combined Training dev and test sets are the official IWSLT 2026 dev and test sets.

2022).

3 Datasets

For our experiments, we used data from the following sources:

1. Official SLT data released by IWSLT organizers
2. AI4Bharat ASR data (Joshi et al., 2025)
3. Additional MT data obtained from Kaggle (dataset link).

The official Bhojpuri–Hindi dataset comprises 22 hours of Bhojpuri news-domain speech and corresponding Hindi translations. Additional ASR data consists of 7 hours of audio speech and corresponding Bhojpuri transcriptions. Additional MT data consists of a total of 44,694 utterances after removing duplicates from the original source (see Table 1).

4 Methodology

4.1 End-to-End

Our end-to-end system combines two complementary ideas: modality adaptation through a dedicated speech-to-text adapter, following M-Adapter (Zhao et al., 2022), and learnable aggregation of intermediate speech encoder layers, following Inter-Connection (Nishikawa and Nakamura, 2023). Concretely, we aggregate selected Wav2Vec2 (Baevski et al., 2020b) encoder’s hidden-layer representations and pass

them through an interconnection adapter that compresses and projects speech embeddings into a form compatible with a pretrained NLLB (Costa-Jussà et al., 2022) decoder.

4.1.1 Speech Encoder

We use the pretrained Bhojpuri-specific Wav2Vec2 backbone from Vakyanish (Chadha et al., 2022). Input speech is converted to mono, resampled to 16 kHz, and fed to the encoder as a raw waveform. Rather than relying only on the final encoder layer, we aggregate hidden representations from layers 6, 8, 10, and 12 using learnable softmax weights. This choice allows the model to exploit complementary information across intermediate and higher-level acoustic representations, while leaving cross-modal alignment to the downstream adapter. The layer selection is motivated by prior observations that different encoder depths capture different linguistic abstractions in self-supervised speech models (Pasad et al., 2024).

4.1.2 Adapter

The adapter is the system’s main alignment module. It first performs temporal compression using strided Conv1d layers with GELU activations (Hendrycks and Gimpel, 2016); in our default setting, two stride-2 convolutions yield an effective 4:1 reduction. This stage shortens the speech sequence to make decoder cross-attention tractable while also introducing a local inductive bias that merges nearby frames into more stable short-range units. The

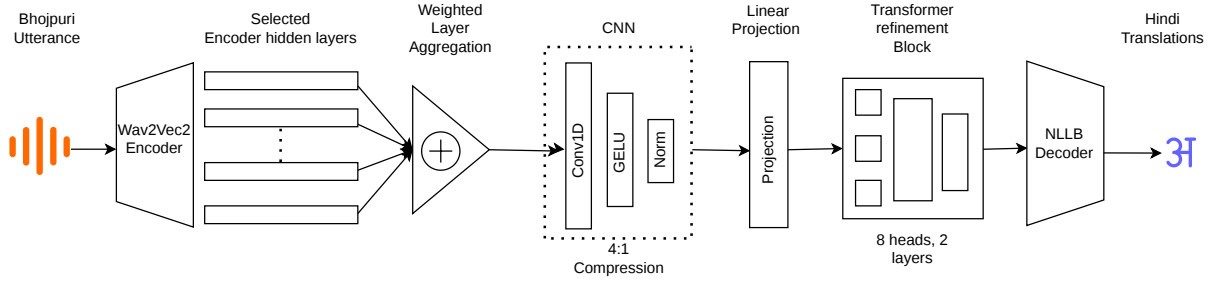


Figure 1: Overview of the utilized modular Bhojpuri-to-Hindi speech translation architecture. A pretrained Wav2Vec2 encoder first converts the input Bhojpuri utterance into frame-level hidden representations from multiple encoder layers. These representations are combined through weighted layer aggregation, compressed by a CNN-based reduction stage, and linearly projected into the hidden space of the NLLB decoder. A lightweight Transformer refinement block then contextualizes the projected speech embeddings before supplying them to the pretrained NLLB decoder as encoder-side context for cross-attention, enabling autoregressive generation of Hindi text.

compressed states are then normalized and linearly projected into the NLLB decoder hidden space. Finally, a lightweight Transformer refinement block (Vaswani et al., 2017) contextualizes these projected speech embeddings before decoding. In the default full-adapter setting, this block uses 2 Transformer layers, 8 attention heads, a feed-forward dimension of $4\times$ the decoder hidden size, and dropout of 0.1. The CNN adapter baseline corresponds to this same pipeline without the Transformer refinement block.

4.1.3 NLLB Decoder

On the text side, we use the decoder of facebook/nllb-200-distilled-600M (Costa-Jussà et al., 2022) as the translation generator. Rather than using the pretrained NLLB text encoder, we bypass it and directly provide the speech-derived adapted states as `encoder_outputs` to the decoder. The decoder then attends to these projected speech representations via encoder-decoder cross-attention and generates Hindi autoregressively, conditioned on the target-language token `hin_Deva`. In this formulation, the decoder contributes multilingual generation and target-side language modeling, while the adapter bears the main burden of mapping speech representations into a form compatible with the decoder’s cross-attention interface.

4.2 Cascade

4.2.1 ASR

For the ASR component, we fine-tune whisper-large-v3 (Radford et al., 2022) on the ai4bharat/RuralWomenBhojpuri dataset, which comprises both real and synthetic speech samples covering a wide range of acoustic variability. Audio is resampled to 16 kHz to match Whisper’s input requirements.

To improve robustness under low-resource and noisy conditions, we apply waveform-level speed perturbation (Ko et al., 2015) and additive Gaussian noise (SNR 15–30 dB), alongside feature-level SpecAugment (Park et al., 2019) for time and frequency masking. For parameter-efficient fine-tuning, we adopt LoRA (Hu et al., 2022) on the Query, Key, Value, and Output attention projections. The model is trained for 10 epochs with a learning rate of 10^{-4} , linear warmup over 5% of the steps, a batch size of 4, and label smoothing of 0.1 (Vaswani et al., 2017). The best dev-set checkpoint is selected, and the learned LoRA parameters are merged into the base model to produce a standalone ASR system that serves as the front-end of our cascaded ASR→MT pipeline.

4.2.2 MT

For the MT component, we fine-tune facebook/nllb-200-1.3B (Costa-Jussà et al., 2022) on a combination of the additional Kaggle dataset (Das) and back-translated data generated from the official training set using the untuned facebook/nllb-200-1.3B. We use cross-

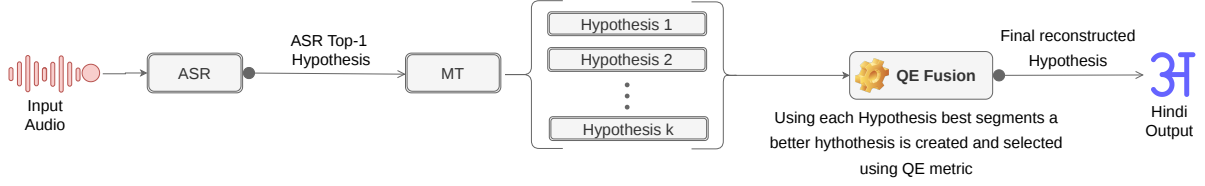


Figure 2: ASR top-1 hypothesis is passed directly to MT, where QE Fusion replaces beam search to reconstruct an alternative hypothesis from k candidates.

entropy as the loss function with the Adafactor optimizer (Shazeer and Stern, 2018), a learning rate of $1e-4$, a gradient-clipping threshold of 1.0, and a batch size of 16. The model was trained for 10 epochs, with the best BLEU scores occurring on Epoch 3.

4.2.3 QE-based Hypothesis Fusion for MT

Neural Machine Translation (NMT) models assign probabilities to translation candidates given a source sentence, and translation errors arise from mismatches between the model’s predictions and the reference/golden translations. Conventional n-gram-based metrics such as BLEU (Papineni et al., 2002) and chrF++ (Popović, 2015) show only weak correlation with human assessments. Consequently, the field has started to turn to neural scoring methods such as COMET (Rei et al., 2020), for better human alignment.

QE Fusion (Vernikos and Popescu-Belis, 2024) uses QE COMET-Kiwi (Rei et al., 2022) scores, which predict translation quality without reference translations, to rerank and reconstruct a superior hypothesis by selecting and combining the best segments across multiple MT hypotheses. Motivated by this, we replace beam search with QE Fusion in our cascade pipeline: the MT component produces 5 hypotheses, which are then scored and reconstructed using a QE model to produce a single refined hypothesis.

5 Experiments

5.1 End-to-End Experiments

These experiments were designed to identify the most effective combination of encoder adaptation, interconnection module, and decoder adaptation for low-resource Bhojpuri-to-Hindi speech translation. Unless stated otherwise, both encoder and decoder backbones were

frozen, the compression ratio of both the CNN adapter and the full adapter was fixed at 4:1, optimization used AdamW (Loshchilov and Hutter, 2017) with batch size 4 and initial learning rate 1×10^{-4} , and learning-rate scheduling followed 5% linear warmup followed by cosine decay. All models were trained on the official IWSLT 2026 Bhojpuri-Hindi data, with augmentation applied only where explicitly stated, and evaluated on the dev set using BLEU (Papineni et al., 2002) and chrF++ (Popović, 2015).

We conducted several preliminary development experiments to validate convergence and guide architectural choices. Since only two systems were officially submitted, we focus the main discussion on those submitted systems and defer the earlier exploratory experiments to the appendix.

We use the CNN adapter experiment as a baseline; this adapter is equivalent to the full adapter without the Transformer refinement block (Figure 1), with the experiment results in Table 2

5.1.1 Adapter with Decoder-side LoRA and Data Augmentation

The third experiment combined the full adapter with decoder-side LoRA and SpecAugment-based augmentation (Park et al., 2019) over the full training set. This was the best-performing configuration overall. Decoding used beam size 5, no-repeat n-gram size 3, and repetition penalty 1.2. Trained on the augmented version of the same 26-hour dataset, this model achieved a peak validation BLEU of 32.77 and a chrF++ of 53.43, showing that stronger modality adaptation, lightweight decoder adaptation, and augmentation were highly complementary in the low-resource setting. This system serves as our PRIMARY submission.

System	BLEU $\uparrow$	chrF++ $\uparrow$
Adapter + Decoder LoRA + Aug.	32.77	53.43
Adapter + Decoder LoRA + Upper Layer Unfreeze + Aug.	30.23	51.56

Table 2: End-to-end system results on the IWSLT 2026 Bhojpuri–Hindi dev set. All models use encoder layers {6, 8, 10, 12} for aggregation and a fixed 4:1 compression ratio.

5.1.2 Adapter with Decoder-side LoRA + Decoder Cross-Attention and Upper Layer Unfreezing with Data Augmentation

The final experiment tested whether broader decoder adaptation could improve upon the best parameter-efficient configuration. Starting from the augmented full-adapter + decoder-side LoRA setup, we additionally unfroze all decoder cross-attention modules and the top two decoder layers (layers 10 and 11 in 0-based indexing), while keeping the remainder of the setup unchanged. Decoding again used beam size 5, no-repeat n-gram size 3, and repetition penalty 1.2. After 40 epochs of training, this model achieved a maximum BLEU score of 30.23 and a chrF++ score of 51.56 on the validation set. Although performance remained strong, it did not surpass the fully parameter-efficient LoRA-based setup; BLEU plateaued around 30 over the final seven epochs, indicating that broader decoder unfreezing increased trainable capacity without yielding further translation gains. This suggests that, in our low-resource setting, preserving pretrained decoder priors was more beneficial than allowing broader decoder adaptation. This system serves as our CONTRASTIVE1 submission.

5.2 Cascade Experiments

For the cascade system, we primarily conducted an experiment to test its effectiveness in the low-resource Bhojpuri→Hindi speech translation setting and to investigate whether errors mainly originate in ASR or MT.

5.2.1 Experiment Setup

Experiments were conducted on the Bhojpuri→Hindi language pair using the IWSLT 2026 official dev set for model selection and the test set for final evaluation. Unless stated otherwise, the following settings were used across the experiments whenever needed: Whisper was run with a decoding temperature of 0.4, batch size of 4, and a no-repeat n-gram

size of 3. Longer audio tracks were segmented into 30-second chunks with a 2-second overlap to handle utterances exceeding the Whisper context limit at inference time. For MT, we used beam search with a beam size of 4 for the baseline and a maximum source and target length of 256 tokens. For QE Fusion, the reference-free quality estimation model Unbabel/wmt22-cometkiwi-da was used with 5 candidates per sentence generated via epsilon sampling, and a reconstruction beam size of $b = 4$ (Vernikos and Popescu-Belis, 2024). All models were evaluated using BLEU, chrF++ and COMET (Rei et al., 2020) for translation quality.

5.2.2 Baseline

For the cascade baseline, the fine-tuned Whisper ASR output was passed directly to the fine-tuned NLLB MT model without any additional setup. The QE Fusion experiment was evaluated against this baseline using BLEU, chrF++, and COMET on the official IWSLT 2026 dev and test sets.

5.2.3 QE Fusion for MT

We replaced standard beam search in the MT block with QE Fusion (Section 4.2.3), generating 5 candidates per sentence via epsilon sampling ($\epsilon = 0.02$, $T = 0.5$) (Vernikos and Popescu-Belis, 2024), which are then scored and reconstructed by Unbabel/wmt22-cometkiwi-da with a reconstruction beam size of $b = 4$ to produce a single refined hypothesis.

As shown in Table 3, QE Fusion yields consistent gains over the beam search baseline across all metrics (BLEU: 15.96→16.09, chrF++: 41.23→41.37, COMET: 0.5394→0.5569), confirming that QE-guided hypothesis reconstruction improves translation quality even in low-resource settings. Table 4 shows a representative example of the 5 MT candidates and the final reconstructed hypothesis. This system serves as our CONTRASTIVE2.

Pipeline	BLEU	chrF++	COMET
<i>Baseline</i>			
ASR → MT	15.96	41.23	0.5394
<i>+ QE at MT</i>			
ASR → MT (QE Fusion)	16.09	41.37	0.5569

Table 3: Cascade results on dev set. Applying QE Fusion at the MT stage yields consistent improvements.

6 Results

We present the results of both end-to-end and cascade systems on the dev and test sets.

6.1 End to End Results

TBW: After results are released

6.2 Cascade Results

TBW: After results are released

7 Discussion

TBW: After results are released

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A Appendix

We define our additional experiments in the following subsections.

A.1 Preliminary End-to-End Experiments

We conducted two preliminary development experiments to validate convergence and to motivate the final submitted systems.

A.1.1 CNN Adapter with Decoder-side LoRA Baseline

The first experiment was a proof-of-concept run to test whether the proposed encoder-adapter-decoder coupling could converge in the low-resource setting. We used the CNN adapter as the interconnection module and applied decoder-side LoRA for lightweight adaptation, without augmentation. Trained for 20 epochs

on approximately 26 hours of data, this system achieved a maximum validation BLEU of 22.01 and a chrF++ of 37.00, establishing the architecture’s basic viability.

A.1.2 Adapter with Decoder-side LoRA without Data Augmentation

The second experiment added a Transformer-based refinement block after the projection stage in order to test whether a more expressive modality-alignment module would improve performance. Decoder-side LoRA was again used for lightweight decoder adaptation, and no augmentation was applied. Because the full adapter introduces additional Transformer capacity, training was extended to 40 epochs. This configuration achieved a maximum BLEU score of 26.14 and a chrF++ score of 42.11, indicating a clear gain over the simpler CNN-based baseline.

A.2 Preliminary Cascade Experiments

A.2.1 LLM as a Post Correction block

We evaluate the LLM-based post-correction block by inserting it between the ASR and MT components. The top-1 ASR hypothesis is passed to the LoRA-fine-tuned ai4bharat/Airavata LLM, which generates a corrected Bhojpuri transcription, which is then fed to the MT model. We evaluate end-to-end using BLEU, chrF++, and COMET against the IWSLT Hindi references.

Introducing the correction block degrades performance significantly (BLEU: 12.03, chrF++: 34.39, COMET: 0.4966). We attribute this to limited Bhojpuri coverage in ai4bharat/Airavata’s pretraining corpus: despite LoRA fine-tuning on 3,200 utterance pairs constructed from Whisper ASR hypotheses paired with the IWSLT 2026 Bhojpuri references, the model frequently hallucinates tokens, switches language mid-sentence, or alters semantic content, compounding the error signal passed to MT. Applying QE Fusion on the LLM hypothesis partially mitigates this degradation (BLEU: 13.02, chrF++: 36.61, COMET: 0.5152), but does not surpass the baseline, confirming that QE-based selection cannot compensate for the LLM’s fundamental coverage gap on Bhojpuri.

Hindi	
MT source	कि हज़ारों वर्षों से हमारी जीवनशैली का हिस्सा रहा है।
Candidate 1	कि हज़ारों वर्षों तक हमारी जीवनशैली का हिस्सा रहा है।
Candidate 2	जो हज़ारों वर्षों से हमारी जीवनशैली का हिस्सा रहा है।
Candidate 3	कि हज़ारों वर्षों तक हमारी जीवनशैली का हिस्सा रहा है।
Candidate 4	कि हज़ारों वर्षों तक हमारी जीवनशैली का हिस्सा रहा है।
Candidate 5	कि हज़ारों वर्षों तक हमारी जीवनशैली का हिस्सा रहा है।
QE selected	जो हज़ारों वर्षों तक हमारी जीवनशैली का हिस्सा रहा है।
Reference	वे हज़ारों वर्षों से हमारी जीवनशैली का हिस्सा रहे हैं

Table 4: QE Fusion example at MT Block